

# MATH 32003

# Number Theory

## Fall 2024

Professor John Bergdall  
[bergdall@uark.edu](mailto:bergdall@uark.edu)

MWF 10:45 - 11:35 AM  
Classroom: SCEN 322

## Overview

This foundational course on numbers focuses on structure and applications. How are numbers atomically built? How do special numbers distribute themselves? Does reorganizing numbers create useful perspectives for solving problems? How are numbers related to secret codes? We are going to ask and hopefully answer questions such as these, and more, all together!

## Learning goals

Learn  
about essential  
problems on  
numbers

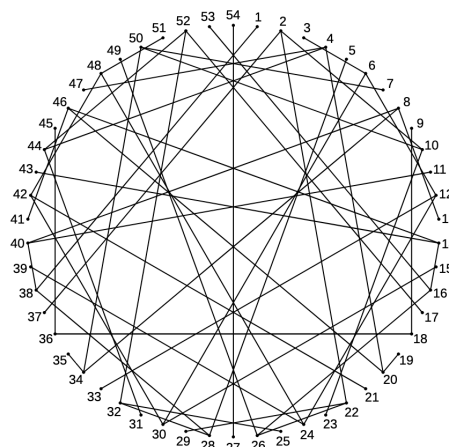
Study  
phenomena with  
congruence  
arithmetic

Explore  
algebraic  
equations and  
congruences

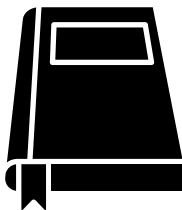
Develop  
applications to  
secret codes

## Play time!

Experiment and play are a major focus of this course. To the right, I was playing around with the dynamics of multiplying by 38 in the world of modular arithmetic modulo 54. What kind of patterns do you see? Symmetries? The questions are endless! You can also play right now, at [demonstrations.wolfram.com/ModularMultiplicationOnACircle](https://demonstrations.wolfram.com/ModularMultiplicationOnACircle/).



# Basic resources



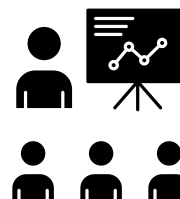
## Textbook

*A Friendly Introduction to Number Theory (4th ed.)* by Silverman  
(Rent/buy: \$70/\$104\*)



## Course materials

Jupyter notebooks on [Google Colab](#)  
(Cost: Free)



## Office hours (SCEN 220)

Mon: 2:00 - 3:00  
Thu: 2:30 - 3:30  
Fri: 9:30 - 10:30

\*Used costs. If costs cause hardship or would limit your course access, email me for support.

# Tasks and grades

Course Tasks are divided into *Supporting Tasks* and *Performance Tasks*. Supporting Tasks keep the class moving. Performance Tasks represent chances to show your understanding of concepts and techniques. **Further details on pages 3-4.**



## Supporting Tasks

Class Participation and Reading Quizzes



## Performance Tasks

Number Theory Labs and Term Project

# Turn-in

**Reading Quizzes** and **Number Theory Labs** are due at the *start of class* (10:45 AM). The **Term Project** will have regular deadlines.

## Revisions

Revisions reinforce your learning. Except for final Project pieces, each Task can be revised *once*.

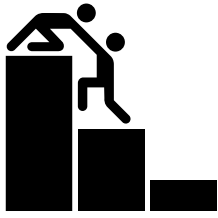
- Revisions for Reading Quizzes are due **Mondays at 11:59 PM**.
- Revisions for Performance Tasks will be due roughly **72 hours** after they are graded and returned.

## Late Supporting Tasks

Completing Tasks by their due dates keeps you on track. Still, you *may turn work in late as part of the revision process*. However, you lose your chance to revise the work. **It is to your benefit to turn in partially completed work** even if it is not perfect.

# Supporting Tasks

**Class participation** gives you voice and agency in your learning. **Reading Quizzes** measure understanding of readings that frame class discussions.



## Class Participation (10 points possible)

- **Attendance.** 1 point for better than 80% attendance.
- **Class surveys.** 1 point for each of 3 surveys during the semester.
- **Class welcomes.** 2 points for each of 3 class welcomes to start class.

**Details.** One way students support learning is structural participation in the course.

- **Attendance.** I expect you to make four out of five meetings. That is all.
- **Class surveys.** At the start, middle, and end of the semester, I survey the course to reinforce what is working and address what is not. The first survey is not anonymous, but the remainder are. You will earn your point by affirming you participated in the survey.
- **Class welcomes.** A class welcome is a short 1-2 paragraph summary of the prior class. You will work with a partner to write the summary, post it to Blackboard, and then read it aloud in the first 2-3 minutes of the next meeting. Sign-ups will be made in the first weeks of class.

## Reading Quizzes (1 point per question)

- **2-5 questions** per reading (roughly **10 quizzes** over semester).
- **Initial responses** in timed and auto-graded quiz on Gradescope.
- **Revisions** will include complete reasoning, in writing.



**Details.** Reading Quizzes questions are *multiple choice* or *select all*.

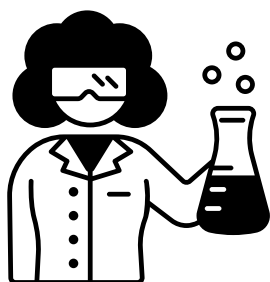
- Before the quiz, you will know a rough form of the questions but not the answer options.
- You take the quiz prior to a class meeting and the questions frame the class discussion.

If you do not get a question initially correct, then revision is possible! Your revision will be writing a detailed response that identifies the correct answer *and* explains the reasoning.

**Revision specifications.** Here is how you will “revise” your Reading Quizzes. **See page 12 for an example.**

- **Multiple choice questions.** Identify the correct answer along with an explanation for why it is correct.
- **Select all questions.** First, identify the correct and incorrect options. Second, explain why the correct ones are correct *and* why the incorrect ones are incorrect.
- Complete and accurate revisions **earn back full points**. Partial explanations may be awarded zero or half a point, based on my discretion.

# Performance Tasks



**Number Theory Labs** are chances to implement explorations based on guided ones done in class. Your **Term Project** will look more deeply at a single number theoretic phenomenon or application.

## Number Theory Labs

- **Focused exploration** with specific formatting and rubric guide.
- **Turned in** electronically through CoCalc.
- **Collaborate!** As long as you attribute credit to your source(s).

**Details.** Our semester will be framed by five (5) labs. The goal of these labs is to make the theoretical content more tangible and to prioritize the *experiential* part of mathematics over the *knowledge* part of mathematics. For each lab, we will dedicate a single day to introducing the concept and beginning our initial work. You will then have roughly one week to complete the lab on your own time. The grading will be done electronically, through CoCalc. **Grading guidelines are on page 5.**

**Revision specifications.** Number Theory Lab revisions are due at the first 11:59 PM on a Mon-Thu no earlier than **72 hours after the initial grades** are returned. (I will announce these deadlines in class and on Blackboard. Do not stress about the complex rule.) To revise your lab, you should copy and then edit the lab notebook and I will re-collect it. I will then assess the re-collected work. *You should not simply explain in writing the mistakes that you made and how to correct them.*

## Term Project

- **Groups of 3-4** working throughout the semester.
- **10-minute presentation** in the final week of class.
- **10-15 page paper** and **Jupyter notebook** due 12/09 at 10:45 AM.



**Details.** The Term Project is a group project that you will work on the entire semester. The purpose is to connect you with some of your interests and expand your ability to independently study number theory and mathematics. Your grade will be based on meeting various deadlines, the initial production of a proposal and outline, the final production of a presentation and paper (supported by a Jupyter notebook), and a testimony to your individual effort on the project. **Grading guidelines are on page 6 and important due dates are on page 10.**

**Revision specifications.** Initial Term Project tasks can be revised. Those revisions are due at the first 11:59 PM on a Mon-Thu no earlier than **72 hours** after the initial grades are returned. The final presentation and paper *cannot* be revised.

# NT Lab Grading

Your Number Theory Labs will be graded using an 10-point rubric. The rubric will be strictly applied because revisions are always possible. **We will go over this rubric during the first week of class.**

**Format.** Your labs will be written as Jupyter notebooks. You will use the formatting capabilities of Markdown to make the notebooks readable and easy to follow. Your notebooks will contain five (5) sections: introduction, data, conjectures, tests, and conclusion. Each section is worth two (2) points.

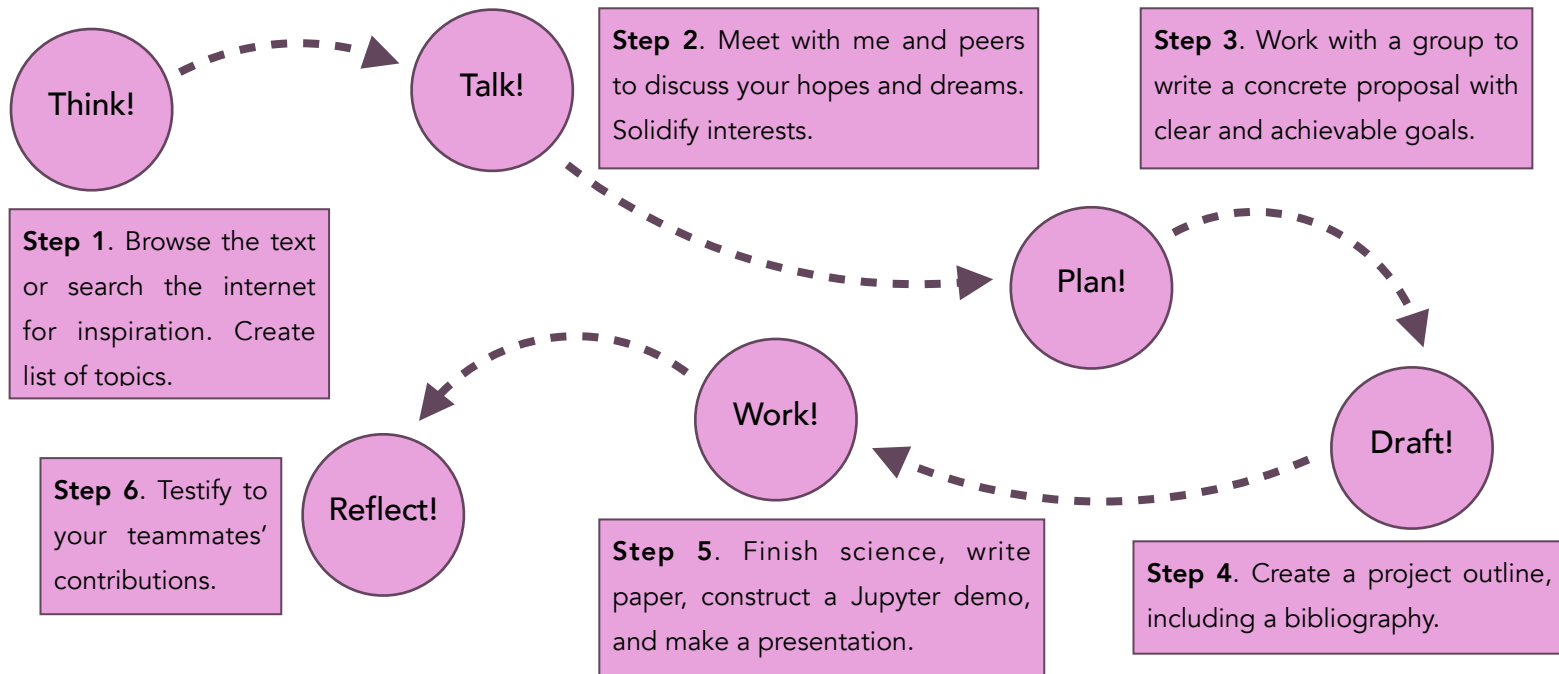
- 1. Introduction.** I will provide a prompt for your lab, of course, but you should still write an introduction *in your own words*. Points will be awarded for describing the context of the exploration as well as an achievable goal. The context may include theoretical background.
- 2. Data.** You should gather data using paper and pencil or the computer. Points will be awarded for accurately gathering data (meaning the data makes conceptual sense and there are no technical mistakes) that is consistent with the stated goal from your introduction.
- 3. Conjecture.** You should formulate conjectures based on your data. Points will be awarded when your conjectures are both consistent with the data and robust. (*Robust* means that there is not a more natural conjecture based on your data.)
- 4. Test.** You are expected to develop tests that *supports* your conjectures. Points will be awarded for your human explanation of the test and for its implementation. The explanation should explain any choices of thresholds.
- 5. Conclusion.** Your conclusion should summarize your findings in the lab. Points will be awarded for the overall positive achievement of the lab and a clear concluding statement. (*Positive achievement* means your conjecture was affirmed by your test, as opposed to contradicted.)

|                                                              |                                                                  |
|--------------------------------------------------------------|------------------------------------------------------------------|
| <b>Intro #1</b><br>Description of context for the lab.       | <b>Intro #2</b><br>Clear statement of an <i>achievable</i> goal. |
| <b>Data #1</b><br>Accurate data gathering.                   | <b>Data #2</b><br>Data gathered is consistent with goal.         |
| <b>Conjecture #1</b><br>The conjecture consistent with data. | <b>Conjecture #2</b><br>The conjecture is robust.                |
| <b>Test #1</b><br>Clear human description of test.           | <b>Test #2</b><br>Successful implementation.                     |
| <b>Conclusion #1</b><br>Overall positive achievement.        | <b>Conclusion #2</b><br>Clear concluding statement.              |

**Caveat.** I will return notebooks that are not clearly formatted. I need to be able to find and discern the five sections and their contents easily. If I do return it, you can always resubmit using the revision process. If you want feedback on the science, you will have to meet with me to discuss.

# Term Project

Your Term Project will happen the entire semester. At each step, you earn points. The total number of points is 30.



## Rubrics

### Steps 1 and 2 — Brainstorming and Discussion (1 point each)

- Graded for completion.

### Step 3 — Project Proposal (2-3 paragraphs, 3 points)

- Is a precise and achievable goal proposed?
- Is there a substantial mathematical element?
- Is there a clear plan for the Jupyter notebook?

### Step 4 — Project Outline (1-2 pages, 3 points)

- Is there a detailed outline of the paper's contents?
- Does the outline match the prior proposal?
- Is there a sufficient bibliography?

### Step 5, Part 1 — Jupyter notebook (5 points)

- Graded using same rubric as Number Theory Labs.

### Step 5, Part 2 — Paper (10-15 pages, double-spaced, 1" margins, 12 pt. font, 8 points)

- |                                                      |                                                      |
|------------------------------------------------------|------------------------------------------------------|
| • Is the paper properly long and formatted?          | • Is there a significant mathematical quality?       |
| • Is there an introduction with a clear thesis?      | • Is there a theoretical component?                  |
| • Is there a clear conclusion justifying the thesis? | • Is there an illustrative example?                  |
| • Does the body of the paper support the thesis?     | • Overall, is the mathematics correct and justified? |

### Step 5, Part 3 — Presentation (8-10 minutes, 6 points)

- |                                                           |                                                      |
|-----------------------------------------------------------|------------------------------------------------------|
| • Is the presentation long enough? Everyone participates? | • Did the presenter explain their main outcome?      |
| • Are visuals and audio crisp and clear?                  | • Did the presenters explain mathematical substance? |
| • Did the presenters motivate their investigation?        | • Did the presenters raise and address questions?    |

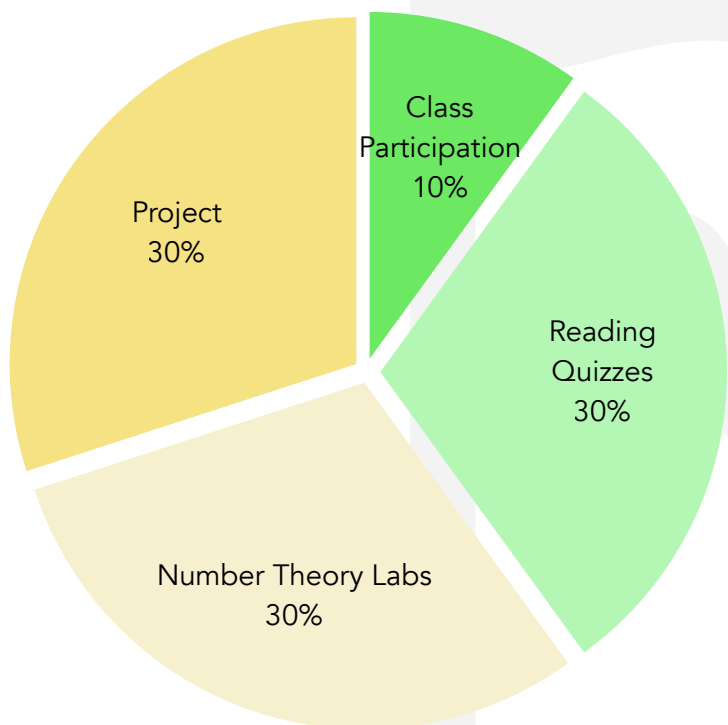
### Step 6 — Contribution testimony (1 page, 3 points)

- How much did you and your teammates contribute the project?

# Final grade

Your Task scores will be weighted according to the percentages in the pie chart below and to the left. Those weighted scores will then determine your final grade according to the table on the right. *I reserve the right to make the gradelines more lenient.*

Task-by-Task weights for final grade



Gradelines

| Grade | Minimum weighted score |
|-------|------------------------|
| A     | 90%                    |
| B     | 80%                    |
| C     | 70%                    |
| D     | 60%                    |

**Further breakdown.** Students sometimes find it difficult to parse the meaning of individual grades aggregated into a final grade.

- *Class Participation.* There are 10 points available for participation, each counts for 1% of your grade.
- *Reading Quizzes.* We anticipate 10 quizzes during the semester, so each will account for 3% of your grade. **Remember to revise your quizzes!**
- *Number Theory Labs.* We anticipate 5 labs during the semester, so each will account for 6% of your grade. **Remember to revise your labs!**
- *Project.* Your project is graded on a 30-point rubric. Each point accounts for 1% of your grade. **Take the rubric seriously.**

# Tips for success

This class is designed for you to pass. The revision process puts you in control of your grade. Everyone will need some help at some point in the semester. So, here are a few tips and resources you can pursue.



## Message!

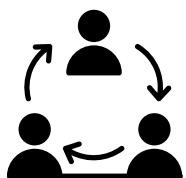
Write for help.



**Ask!** Me early and often.

## Tips for the stuck and frustrated.

- Take a break. Walk away.
- Explain to someone why you are stuck.
- Check your assumptions on a problem.
- Work out a simpler example.
- Search the internet for key terms.
- Keep going!

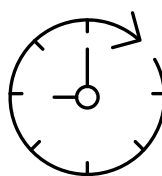


## Collaborate!

With peers.



**Read!** The text is a guide.



**Budget!** Take your time.

# Further Resources

## Appointments

Office hours will not always fit your schedule. Email me to set appointments. In your initial message, please explain (i) what you want to talk about, (ii) *your* availability and (iii) if meeting in person is important (Zoom is possible).



## Email

Your questions are important. Email me or post to Blackboard. (I may ask you to do this anyways.) Responses may take 24 hours, but send a reminder if I do not reply.

## Access

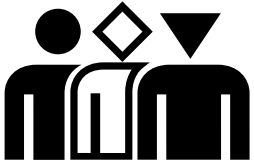
I want you to pass this class, with success. If you find resources not being provided, tell me and I will work with you to fix the issue. For some students, the Center for Educational Access ([cea.uark.edu](http://cea.uark.edu)) coordinates student accommodation requests. Please request your accommodation letter early semester and meet with me to discuss it.



# Class formation

## Names and pronouns

You deserve to be called how you want. Let me know your preferred name and pronouns any time. When meeting one another, re-introduce yourself *frequently*. Everyone's names, pronouns, etc. should be properly used.



## Diversity

I pledge attention to your identity and experience, regardless of age, background, beliefs, ethnicity, gender identity and expression, national origin, racial identity, religious beliefs, sexual orientation, and any other visible or non-visible categories. You are expected to do the same for me and your peers.

## Academic integrity

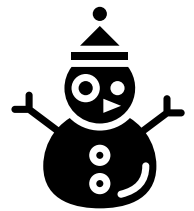
I trust my students to follow (i) the University of Arkansas Academic Integrity Policy and (ii) my own explicit assignment instructions. When you have questions about a boundary or rule, ask me rather than assuming you know the answer.



# Technical policies

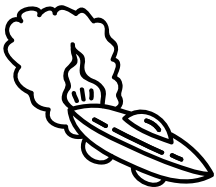
## Inclement weather

If the University closes due to inclement weather, our course will meet online. The meeting will be recorded, as dictated by University policy. Any assessment or due date will be postponed until the next day the University is open.



## Class cancellation (emergencies)

If class is cancelled on short notice, I will work with the math office to try to find a substitute. If a substitute is found, class will be held. If not, alternative learning materials and assessments will be supplied. Regardless, an announcement will be sent promptly via email and posted to Blackboard.



## Devices

I will record class meetings using in-room recording systems. You will often use your devices (phone, tablet, computer, etc.) to participate in meetings. You are expected to use those devices productively. Non-course activity distracts your peers from learning and me from teaching. It will not be tolerated.



# Schedule

| "Unit" | Dates         | Theory Topic                                                  | Reading (Silverman)  |
|--------|---------------|---------------------------------------------------------------|----------------------|
| 0      | Aug 19-23     | Course welcome, introduction to Jupyter notebooks and CoCalc. | Ch. 1                |
| 1      | Aug 26-Sep 11 | A sample of basic problems in Number Theory.                  | Ch. 2-3, 5-7, 12-13  |
| 2      | Sep 13-Sep 27 | Congruence arithmetic.                                        | Ch. 8-11, 16-17      |
| 3      | Sep 30-Oct 16 | Algebraic equations and quadratic residues.                   | Ch. 20-21, 28-29     |
| 4      | Oct 18-Nov 1  | Quadratic reciprocity and beyond.                             | Ch. 22-25            |
| 5      | Nov 4-Nov 25* | Applications of congruence arithmetic to cryptography.        | Ch. 18 + supplements |
| 6      | Dec 2-Dec 4   | Project Presentations                                         | —                    |

\*From November 4- November 25, each Monday will be a working day for the Term Project.

# Project due dates

| Step # | Date             | Due                          |
|--------|------------------|------------------------------|
| 1      | Sep 6            | Topic brainstorm             |
| 2      | Sep 16-20        | Topic meeting and discussion |
| 3      | Oct 11           | Project proposal             |
| 4      | Nov 1            | Project outline              |
| 5      | Dec 2-Dec 4      | Presentation                 |
| 6      | Dec 4            | Contribution testimony       |
| 5      | Dec 9 @ 12:15 PM | Paper and Jupyter notebook   |

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## Reading Quiz Example

Revising *select all that apply* questions requires explaining both the logic behind the “right” answers and the logic behind the “wrong” answer. Let us see how that looks in practice.

**Question. (Select all that apply)** Which of the following indefinite integrals are calculated accurately?

(a)  $\int \cos(x) dx = \sin(x) + C.$

(b)  $\int \sin(x) dx = \cos(x) + C.$

(c)  $\int \tan(x) dx = \sec^2(x) + C.$

**Sample “revision”.** The integral (a) is accurate but (b) and (c) are not. The answer (a) is true because  $\cos(x)$  is the derivative of  $\sin(x)$  and so  $\sin(x)$  is the anti-derivative of  $\cos(x)$ . Next, (b) is false because it should say  $\int \sin(x) dx = -\cos(x) + C$ . And, (c) is false because, in fact,

$$\int \tan(x) dx = \ln(|\sec(x)|) + C.$$

**Tip.** To show something is *false*, you can provide a corrected statement.

**Question. (Select all that apply)** Suppose  $y = f(x)$  and  $y = g(x)$  are two curves over  $x$  lying in  $[a, b]$ . Which of the following statements are accurate?

(a) The area between the two curves is always non-negative.

(b) The integral  $\int_a^b f(x) - g(x) dx$  represents the area between the two curves.

(c) The integer  $\int_a^b f(x) - g(x) dx$  is always non-negative.

(d) The integral  $\int_a^b |f(x) - g(x)| dx$  represents the area between the two curves.

**Sample “revision”.** The answers (a) and (d) are accurate, while (b) and (c) are not. For (a), areas are always non-negative so that is easy. For (d), this integral is how you calculate the area (we learned this in the reading). The answer (c) and (b) are not true. For instance suppose  $f(x) = x$  and  $g(x) = 2$  over the interval  $[0, 1]$ . Then

$$\int_0^1 x - 2 dx = \frac{1}{2} - 2 = -1.5.$$

So, (c) fails. The main issue is  $g(x) = 2$  lies above  $f(x) = x$  on  $[0, 1]$  and so  $\int_0^1 f(x) - g(x) dx$  calculates the *negative* of the area in this case. This explains why (b) is false.

**Tip:** To show something is *false*, you can also provide a counter-example like we do here for cases (b) and (c).